# PromoFinder

Problem 1 · Find the Promos · solution note

Tested on all 50 episodes of "I Married a Broken Soldier. He Was a Duke"

**Rajesh Choudhary**
Roll no. 23117109
Mechanical Engineering, IIT Roorkee
rajesh_c@me.iitr.ac.in

PromoFinder reads the screenplays of a whole microdrama and returns a ranked slate of 20–60 s promo clips. Each clip comes with its episode, the start and end of every segment, the hook line, the cliffhanger line and a one-line reason, and every timestamp and quote is checked against the screenplay.

**50 → 387 → 12**
episodes, candidate clips, recommended slate

**8/8 · 7/7**
human-picked clips found (development sets)

**0 / 387**
shown quotes outside their clip (26/376 before the strict validator)

**70 / 70**
automated checks passed: core, formats, adapter, browser

**Post these first** (top of the 12-clip slate: each opens on a hook inside 3 seconds and stops on an unresolved moment)

**#1** Ep36 02:26–03:16 · 50s
**"What wedding?"**
**CUTS ON** ...their dead Duke is coming home.
*Why:* Brilliant emotional and narrative arc from learning about the wedding and execution to...

**#2** Ep43 00:00–00:21 · 21s
**"You're lying!"**
**CUTS ON** No...
*Why:* Great escalation from the guards refusing arrest to the reading of the treasonous...

**#3** Ep23 01:19–02:11 · 52s
**"Someone is poisoning her."**
**CUTS ON** Loud, metallic clack of the door latch turning, followed by the slow...
*Why:* The ultimate master promo: opens with the jaw-dropping arsenic revelation and builds...

**The problem (Kuku, Problem 1)**
An editor watches every episode to pick promo moments by hand, and new shows arrive weekly. A promo is a 20–60 s clip that opens on a hook in the first 3 seconds, ends on an unresolved cliffhanger and makes sense to someone who has never seen the show. The task: find them, give episode and start/end of every segment, rank them, and check where the tool picks badly. Text only: no editing, music or overlays.

**What PromoFinder does**
Reads every episode, builds candidate windows, has a model judge them, makes every timestamp and quote verifiable, ranks the best clips of the whole show head to head, and hands over a slate of 12 plus all 387 ranked candidates. A dashboard lets an editor preview, shortlist, export and run it on a new show. The slate holds at most two clips per episode (12 episodes appear).

## 01 Architecture and workflow

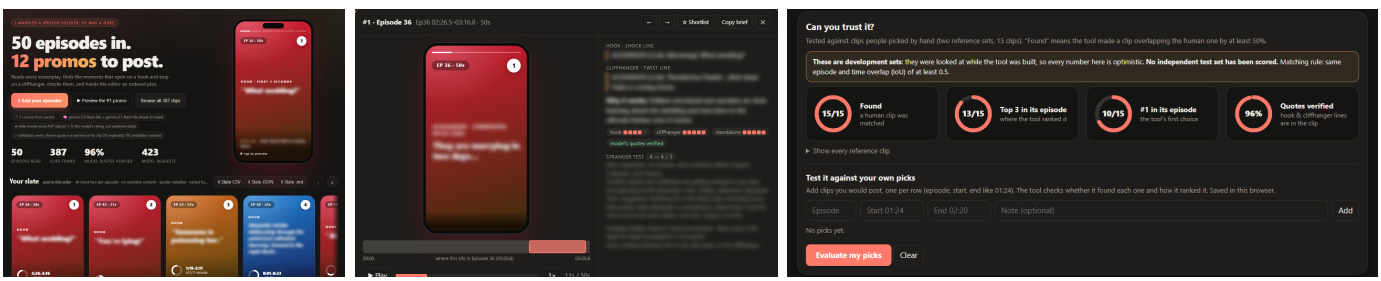
1 Parse → 2 Funnel → 3 Judge → 4 Stranger test → 5 Validate → 6 Rank → 7 Slate → 8 Editor

plain code · AI model · code + AI · human

| | | |
|---|---|---|
| 1·2 | code | Parse any supported format; score every 20–60 s window on beat boundaries; keep 12 per episode. |
| 3 | AI, 1 call per episode | Reads the episode, quotes hook and cliffhanger, rates them 1–5, writes the reason, may add windows the rules missed. |
| 4 | AI, finalists only | Stranger test: does someone who has seen nothing follow the clip? Tries a setup segment if not. |
| 5 | code (strict) | Snaps to beats, enforces 20–60 s, dedupes, makes every quoted line a real line of its clip. |
| 6·7 | AI + code | The 60 best clips of the whole show (at most 2 per episode) meet head to head; code picks a varied slate and trims it to open on the hook. |
| 8 | human | An editor accepts, reorders or drops in the dashboard; every clip has a clip_id for tracking. |

**Technical decisions that matter**

- **Code decides, the model proposes.** Timestamps, quotes, length, duplicates and sensitive-content holds are checked by code; a model line that fails is pulled in, replaced or rejected, never trusted.
- **Head to head, not absolute scores.** Model 1–5 ratings bunch up across episodes, so the finalists are compared in pairs, each pair in both orders.
- **Not tied to one show.** Generic word lists, five input formats, swappable model, and a quality report that needs no human labels.




Real screenshots of the working app: dashboard and slate, the video-style preview, and the evaluation view (script text blurred).

## 02 How it was evaluated

**Setup.** The tool was run on all 50 episodes of the pack. Two reference sets exist: set A, 8 clips I picked by hand while the tool was being built, and set B, 7 clips drafted by the AI assistant from the scripts. **Both are development sets:** I looked at them while designing the tool, so every number below is optimistic, and **no independent test set has been scored**. Both files are frozen (SHA-256) and labelled so; a fresh, boundary-checked set can be scored with one command. **Matching rule:** same episode and time-span IoU of at least 0.5, because human and tool boundaries rarely coincide (only 4 of A's and 1 of B's picks sit on exact screenplay beat boundaries). Sets are tiny: one clip moves a recall figure by 12–14 points.

| Measure | Result (measured on this run) |
|---|---|
| Human clips found among all clips | A: **8/8** · B: **7/7** |
| Inside the overall top 10 / 12 / 20 (A) | 12% / 12% / 50% |
| Inside the overall top 10 / 12 / 20 (B) | 29% / 43% / 43% |
| Best clip of its own episode: #1 / top 3 | A: 50% / 88% · B: 86% / 86% |
| MRR / median rank of the best matching clip | A: 0.073 / 25.0 · B: 0.088 / 31 |
| Slate clips that match a human pick (precision, a lower bound) | A: 1/12 · B: 3/12 |
| Shown hook/cliffhanger line outside its clip | **0/387** (model's 774 quoted lines: 666 right, 92 pulled in, 16 replaced) |
| Hook within 3 s / clip ends on its cliffhanger | all clips: 99% / 82% · slate: 100% / 100% |
| Head-to-head judge: pairs split between the two orders | 21% (a second model 20%; the two models agree, Spearman 0.76) |

**Failure analysis**

- **Cross-episode ranking is unreliable.** With this show's own words in the funnel 3 of 8 human picks reached the top 10; after I made the funnel generic it was 1 of 8. I treat that swing as noise and claim no improvement.
- **Inside an episode, set A:** the tool's top clip is my pick in 4 of 8 episodes. In Ep 3 and Ep 18 its top clip is a different scene (0–7% overlap with mine); in Ep 41 and Ep 42 it is a neighbouring cut (38%). My picks there are its 2nd or 3rd clip.
- **Ep 3 (the stranger's proposal)** is found but ranks 291 of 387: the model rates its cliffhanger 1/5 because the scene resolves.
- The spoiler check cannot see that the show's title gives the twist away; the model's "why" texts praise almost everything.

**Limitations and assumptions**

- Evaluation is on development sets only; model scores are model judgements, not audience behaviour.
- Of the ten dimensions the brief suggests, the model rates hook, cliffhanger and standalone clarity; a newcomer test, a spoiler label and duration add three more; emotional intensity, curiosity, retention and social fit are not scored separately.
- Text only; no video, audio or viewer data. Word lists are English. Clips with estimated times (scripts without timestamps) are flagged as approximate.
- A viewer who clicks a promo is assumed to start at Episode 1 (a hypothesis).

## 03 Scaling, tools, and business impact

**Scaling to many dramas**

- 423 model requests for a 50-episode show (about 30 minutes at the free tier's 15 calls a minute); cached steps are free, so a rerun costs nothing. At 250 shows a month that is about 110,000 requests: use a paid or batch endpoint.
- No source change per show: folder or ZIP upload, five input formats, a model and endpoint set in one config file, per-stage rate limits, and a quality report for shows nobody has labelled.
- Stress-tested on a disguised copy (new names, odd file names, missing headers, no timestamps, inline times, .srt).

**Tools and one AI mistake**

Claude Code (planning, code, tests), Gemini 3.5 Flash-Lite and 3.1 Flash-Lite (runtime, free tier), Playwright (browser tests), ReportLab (this PDF). **Mistake:** the AI and I reported "top-10 recall 1/8 → 3/8" as an improvement; removing this show's own words from the funnel showed it was noise and partly fitted to one show.

**Does it help? A real experiment**

Post the slate (arm A) and the editor's usual picks or random clips (arm B) on the same channels in the same weeks; tag each link with the clip's **clip_id** (in every CSV export).

| | |
|---|---|
| 3-second hold, completion | A ≥ B (Kuku's benchmark 50–60%) |
| Installs per 1,000 views | main metric: A beats B by a margin set in advance |
| Ep 1 started; Ep 1 → 3 retention | guardrail: no clickbait |
| Editor keeps / drops | tells us which rule to fix |

Proposed decision rules, not results: I have no Kuku audience data.

**Prototype access.** Repository: link in the submission form. **python app.py --scripts <folder>** opens the dashboard at http://127.0.0.1:8000 (upload files or a ZIP, Run, preview, export); **python run_pipeline.py --input <folder>** runs the same pipeline; **python eval/run_eval.py** and **python eval/run_all_tests.py** reproduce the evaluation and tests. Raw outputs of the full 50-episode run: the out/ folder, shared by a restricted link. The input scripts are confidential and are not in the repository.